Calculate the control limits for mean chart and the range chart and then comment on the state of control. (conversion factors for $n = 5,\ A_2 = 0.58\ , D_3 = 0$ and $D_4 = 2.115$)

15. Ten samples each of size five are drawn at regular intervals from a manufacturing process. The sample means ($\bar{X}$) and their ranges (R) are given below:

Sample number	1	2	3	4	5	6	7	8	9	10
$\bar{X}$	49	45	48	53	39	47	46	39	51	45
R	7	5	7	9	5	8	8	6	7	6

Calculate the control limits in respect of $\bar{X}$ chart. (Given $A_2 = 0.58\ , D_3 = 0$ and $D_4 = 2.115$) Comment on the state of control.

16. Construct $\bar{X}$ and R charts for the following data:

Sample Number	Observations		
1	32	36	42
2	28	32	40
3	39	52	28
4	50	42	31
5	42	45	34
6	50	29	21
7	44	52	35
8	22	35	44

(Given for $n = 3,\ A_2 = 0.58\ , D_3 = 0$ and $D_4 = 2.115$)

17. The following data show the values of sample mean ($\bar{X}$) and its range (R) for the samples of size five each. Calculate the values for control limits for mean , range chart and determine whether the process is in control.

Sample Number	1	2	3	4	5	6	7	8	9	10
Mean	11.2	11.8	10.8	11.6	11.0	9.6	10.4	9.6	10.6	10.0
Range	7	4	8	5	7	4	8	4	7	9

(conversion factors for $n = 5,\ A_2 = 0.58\ , D_3 = 0$ and $D_4 = 2.115$)

18. A quality control inspector has taken ten samples of size four packets each from a potato chips company.The contents of the sample are given below, Calculate the control limits for mean and range chart.

XII Std - Business Maths & Stat EM Chapter 9.indd 227 2/28/2020 5:11:34 PM